

Jeniya Sultana

Curriculum Vitae

The University of Texas at El Paso, El Paso, Texas

jultana2@miners.utep.edu | [linkedin.com/in/jeniyasultana](https://www.linkedin.com/in/jeniyasultana) | [Google Scholar](https://scholar.google.com/citations?user=jultana2) | github.com/jeniya1378 |

jeniya1378.github.io/Jeniya_Sultana.github.io

Research Interests

Wearable physiological sensing (EEG, EMG, PPG) for construction worker safety and well-being; mental fatigue and cognitive load assessment; neuroergonomics and human factors of wearable robotics and exoskeletons; machine learning under limited and noisy data, including generative data augmentation; research translation and technology adoption in construction.

Education

Ph.D., Civil, Environmental and Construction Engineering

The University of Texas at El Paso

Jan 2025 – Present

El Paso, TX

Research focus: wearable sensing and machine learning for construction safety

M.S., Computer Science

Missouri State University

Jul 2024

Springfield, MO

Thesis: *Towards Dynamic Context Detection from Voice Commands and Conversations with Smart Assistants in Smart Homes*

Graduate Certificate, Software Engineering

Missouri State University

May 2024

Springfield, MO

B.Sc., Computer Science and Engineering

Chittagong University of Engineering and Technology (CUET)

Dec 2019

Chittagong, Bangladesh

Undergraduate thesis recognized among the best in the department

Research Experience

Graduate Research Associate

The University of Texas at El Paso

Jan 2025 – Present

El Paso, TX

- Design and conduct human-subjects experiments collecting EEG, EMG and PPG signals during construction tasks and awkward working postures.
- Develop signal processing and machine learning pipelines to assess mental fatigue, cognitive load and physical strain.
- Apply generative adversarial networks (GANs) to augment small, noisy biosignal datasets and improve model generalization across participants.
- Evaluate the physical and cognitive effects of passive exoskeletons using multimodal physiological monitoring (Construction Research Congress 2026; HFES Regional Conference 2026 Best Poster Award).
- Prepare experimental protocols and human-subjects documentation, and collaborate with faculty and lab members on manuscripts and presentations.

Entrepreneurial Lead, NSF Innovation Corps (I-Corps)

National Science Foundation

Aug 2025 – Dec 2025

El Paso, TX

- Selected as graduate Entrepreneurial Lead to explore translation of lab research on wearable sensing toward real-world adoption.
- Led customer discovery, conducting **50+ interviews** with industry stakeholders to test assumptions and validate problem-solution fit.
- Synthesized interview findings with faculty and industry mentors to assess feasibility, market need and adoption pathways.
- Team received competitive NSF I-Corps funding for research translation and customer discovery.

Graduate Research Assistant

Aug 2022 – Jul 2024

Department of Computer Science, Missouri State University

Springfield, MO

- Developed a context recognition framework for short spoken text using transfer learning, SBERT encoding and attention (IEEE CCNC 2024).
- Built DyCoDe, a dynamic context detection framework using clustering and zero-shot classification so smart home systems adapt to new environments (IEEE CCNC 2025).
- Designed ten metamorphic relations to verify and validate neural network context recognition models (IEEE CIoT 2024).
- Awarded Graduate College thesis funding for the M.S. thesis proposal.

Publications

Conference Proceedings

- [1] **J. Sultana**, A. A. Raheem, J. Jang, J. S. Lee, Y. Kim, J. Kim, J. Kim, “EEG-Based Mental Fatigue Assessment during Exoskeleton-Assisted Construction Postures: A GAN-Augmented Approach,” in *Construction Research Congress 2026*, ASCE, pp. 349–358, 2026. doi: 10.1061/9780784486962.034
- [2] **J. Sultana** and R. Iqbal, “DyCoDe: Dynamic Context Detection from Voice Commands and Conversations in Smart Homes,” in *Proc. IEEE 22nd Consumer Communications & Networking Conference (CCNC)*, 2025.
- [3] F. H. Athina, **J. Sultana**, D. D. Spandan and R. Iqbal, “Metamorphic Testing for Investigation of Context Recognition from Smart Home Voice Commands,” in *Proc. 7th Conference on Cloud and Internet of Things (CIoT)*, 2024.
- [4] **J. Sultana** and R. Iqbal, “A Framework for Context Recognition from Voice Commands and Conversations with Smart Assistants,” in *Proc. IEEE 21st Consumer Communications & Networking Conference (CCNC)*, pp. 218–221, 2024.

Book Chapters

- [5] A. Mondol, **J. Sultana**, M. Ahmed and A. B. Rashid, “Misinformation Detection in Cyber Smart Cities,” in *Cybersecurity for Smart Cities: Practices and Challenges*, Springer, pp. 105–120, 2023.

Thesis

- [6] **J. Sultana**, “Towards Dynamic Context Detection from Voice Commands and Conversations with Smart Assistants in Smart Homes,” M.S. Thesis, Missouri State University, 2024.

Presentations and Posters

J. Sultana et al., “Neuroergonomic Evaluation of Exoskeleton-Assisted Construction Postures Using Multimodal Physiological Monitoring,” poster, Human Factors and Ergonomics Society (HFES) Regional Conference, El Paso, TX. <i>Best Poster Presentation Award</i> .	Apr 2026
EEG-based mental fatigue assessment with GAN augmentation, Construction Research Congress (ASCE)	2026
Dynamic context detection (DyCoDe), IEEE Consumer Communications & Networking Conference	2025
Metamorphic testing for context recognition, Conference on Cloud and Internet of Things	2024
Context recognition framework for smart assistants, IEEE Consumer Communications & Networking Conference	2024
Research presentations, 30th and 31st Frank Einhellig Graduate Interdisciplinary Forum, Missouri State University	2023, 2024
Research presentation, CSC-703 Research Methods, Missouri State University	

Honors and Awards

Best Poster Presentation Award , HFES Regional Conference, El Paso, TX	Apr 2026
NSF I-Corps Team Funding , National Science Foundation, research translation and customer discovery	2025
Bears Spotlight , featured by the Department of Computer Science, Missouri State University	2024
Graduate College Thesis Fund , Missouri State University, for proposal clarity and justification	Fall 2023
Best Undergraduate Thesis recognition, Department of CSE, CUET	2019

Second Runner-Up (female category), National Collegiate Programming Contest (NCPC), Bangladesh

Professional Service and Leadership

Session Chair , Human Factors and Ergonomics Society (HFES) Regional Conference, El Paso, TX	Apr 2026
Officer , Institute of Transportation Engineers (ITE) Student Chapter, The University of Texas at El Paso	2025 – 2026
Organizer , ROSE teacher research symposium, Missouri State University	2023
Organizer , computer science Olympiads for K–12 students, CODERS Project, Missouri State University	2022 – 2024

Teaching Experience

Lecturer <i>Bangladesh University of Textiles</i>	Nov 2021 – Aug 2022 <i>Dhaka, Bangladesh</i>
• Taught computer programming courses to undergraduate students across multiple textile engineering departments. • Managed lesson planning, examinations, grading and attendance.	
Private Tutor <i>Study Zone</i>	2015 – 2022 <i>Chittagong, Bangladesh</i>
• Taught mathematics, ICT and chemistry to 12th-grade students; provided peer tutoring to undergraduates in mathematics and computing.	

STEM Education and Outreach

STEM Instructor <i>Discovery Center of Springfield</i>	Oct 2024 – Dec 2024 <i>Springfield, MO</i>
• Developed hands-on programs aligned with national STEM standards for K–12 students and teachers. • Taught in-house, at schools and community events, and through nationwide distance learning; led science summer camps.	
Graduate Assistant, CODERS Project <i>Computer Science Opportunities, Development, and Education in Rural Schools, Missouri State University</i>	Aug 2022 – Jul 2024 <i>Springfield, MO</i>
• Developed curriculum and lesson plans to teach K–12 teachers fundamental computer science concepts. • Led hands-on workshops with block coding, Micro:bit, Cutebot, Scratch, robot dogs, smart home kits and drones. • Organized computer science Olympiads and refined instructional strategies with classroom teachers.	
Graduate Assistant, ROSE Project <i>Research Opportunity for Smart Environments, Missouri State University</i>	Summer 2023 <i>Springfield, MO</i>
• Taught middle and high school STEM teachers computational thinking, Python and introductory machine learning. • Mentored teachers through research projects from literature review to evaluation, and supported poster preparation.	

Selected Projects

• Students’ attention monitoring: classified attention states from facial landmarks (MediaPipe FaceMesh, Xception) on the DAiSEE dataset. • Research paper classification: neural classifier using TF-IDF and feature selection on titles and abstracts. • Thematic and sentiment analysis tool: end-to-end text analysis tool designed for non-technical users. • Tracking humans carrying baggage: hybrid tracker combining Mean Shift, particle filter and sliding window. • Crude oil price prediction: LSTM time-series forecasting on prices from 1987 to 2019. • Web systems: school, library and employee management systems; bus ticket reservation; food ordering with Google location APIs (PHP, MySQL, JavaScript).	
---	--

Certifications and Training

• Lean Six Sigma: White Belt, Yellow Belt, Green Belt • NSF I-Corps: customer discovery and research translation, National Science Foundation (2025)	
--	--

- **CITI Program:** Social-Behavioral-Educational Researchers, Human Subjects Research
- **Web Strategy and Development,** Missouri State University
- **CCNA,** Institute of Information and Communication Technology (IICT), CUET
- **Web Development** (IICT, CUET and Udemy); **App Development** and **Ethical Hacking** (Udemy); **Graphic Design** (Brain Tuners)
- **Industrial Attachment,** Bangladesh Telecommunications and Business Services (BBTS), 2019

Technical Skills

- **Machine learning:** deep learning, GANs, transformers and SBERT, attention models, LSTM, transfer learning, zero-shot classification, clustering
- **Sensing and signals:** EEG, EMG and PPG acquisition and processing; multimodal physiological monitoring; experimental protocol design
- **Programming:** Python, MATLAB, C, C++, Java, SQL, JavaScript, Node.js, Express.js, PHP, HTML/CSS
- **Tools:** OpenCV, MediaPipe, JupyterLab, VS Code, Weka, MongoDB, MySQL, Android Studio, Packet Tracer, Adobe Photoshop and Illustrator

References

Dr. Razib Iqbal

Associate Professor, Department of Computer Science,
Missouri State University
RIqbal@MissouriState.edu
417-836-4944

Dr. Lloyd A. Smith

Professor, Department of Computer Science, Missouri State
University
LloydSmith@MissouriState.edu
417-836-4834

Dr. Mohammed Y. Belkhouche

Assistant Professor, Department of Computer Science,
Missouri State University
YassineBelkhouche@MissouriState.edu
417-836-5789

Dr. Songfeng Zheng

Professor, Department of Mathematics, Missouri State
University
SongfengZheng@MissouriState.edu
417-836-6037